Unlocking Writing Style with Stylometry

Multi-Genre Text Classification Based on Writing Style

TEAM DETAILS :

1. G.Yeshwanth Ram – SE23UARI135(Team Leader)
2. A.Saketh Krishna – SE23UARI013
3. Bandi Adrija – SE23UARI004
4. L.Mohan – SE23UARI142
5. V.Dheeraj – SE23UARI163

Git-Repo : https://github.com/Yeshwanth110905/NLP-Project
Web-site : https://b50448c5e6e6d6be06.gradio.live/

What is Stylometry?

The Science of Style

Stylometry is the scientific study of linguistic style, focusing on **how** texts are written rather than **what** they're about.

It analyses patterns in word choice, sentence structure, and punctuation to reveal unique writing signatures.

The Challenge

Most NLP systems classify by topic—sports, politics, technology. Few models focus on style-based classification.

We need: A robust classifier that differentiates writing styles independent of content.

Why it matters: Literary analysis, authorship attribution, educational feedback, and content recommendation.

Objective & Dataset

Normal
Everyday conversational

Poetic
Figurative, rhythmic

Narrative
Story-telling

Journalistic
News reporting

Academic
Formal research

Philosophical
Abstract essays

Dataset: Custom multi-genre corpus with preprocessed documents (Unicode normalisation, minimum 5 words). Split: 70% train / 10% validation / 20% test.

Our Hybrid Approach

A sophisticated two-stream architecture combining handcrafted features with deep learning

Stylometric Features

40+ handcrafted linguistic indicators capturing explicit style patterns

Semantic Embeddings

384-dimensional vectors from pre-trained transformers capturing implicit meaning

Fusion Network

Neural network combines both streams for optimal style classification

Methodology: Hybrid Approach

Dual-Stream Feature Extraction

Stream 1: Stylometric Features

40+ features capturing:

- **Lexical:** Word length, vocabulary richness (Type-Token Ratio, Yule's K)
- **Syntactic:** Sentence structure, POS tags, parse depth, passive voice
- **Punctuation:** Comma, quotes, exclamation patterns
- **Readability:** Flesch Reading Ease, Flesch-Kincaid Grade
- **Style Markers:** First-person pronouns, dialogue indicators

Stream 2: Semantic Embeddings

SentenceTransformer (paraphrase-multilingual-MiniLM-L12-v2) generates 384-dimensional dense vectors.

Captures semantic content and contextual meaning alongside stylistic patterns.

Model Architecture

1 **Input Layer**

Concatenated features (424 dimensions)

2 **Batch Normalisation**

Stabilises learning process

3 **Dense Layer 1**

512 units + ReLU + Dropout (0.3)

4 **Dense Layer 2**

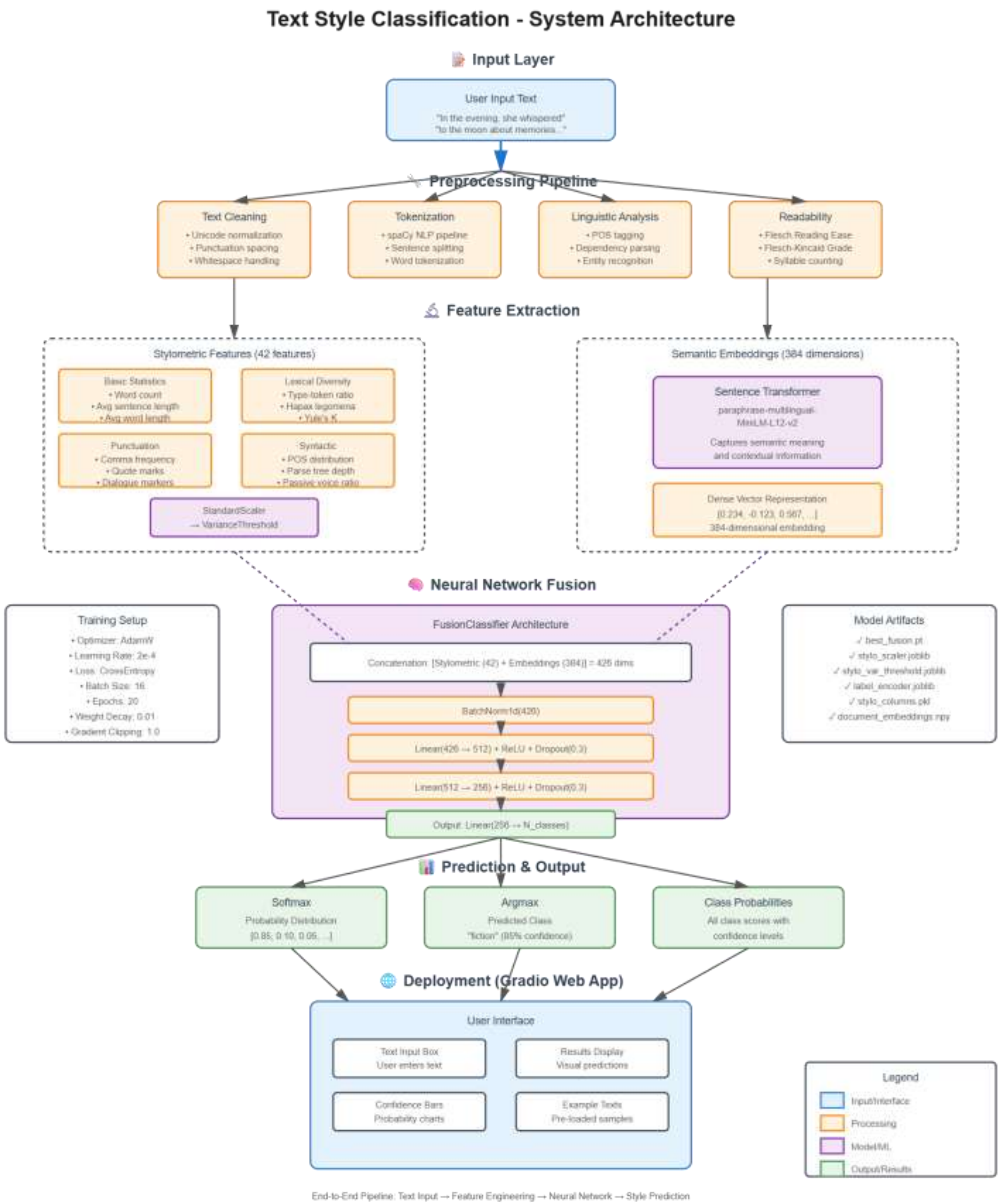
256 units + ReLU + Dropout (0.3)

5 **Output Layer**

7 classes + Softmax

Training Configuration

- Optimiser: AdamW (lr=2e-4, weight_decay=0.01)
- Loss: Cross-Entropy
- Batch size: 16
- Epochs: 100



	precision	recall	f1-score	support
Academic	0.95	1.00	0.98	20
Normal	1.00	1.00	1.00	53
journalistic	1.00	0.67	0.80	6
narrative	0.91	1.00	0.95	21
philosophical	1.00	1.00	1.00	11
poetic	1.00	0.95	0.97	20
accuracy			0.98	131
macro avg	0.98	0.94	0.95	131
weighted avg	0.98	0.98	0.98	131

Results: Strong Performance

98%

Accuracy

Test set performance

95%

Macro F1 Score

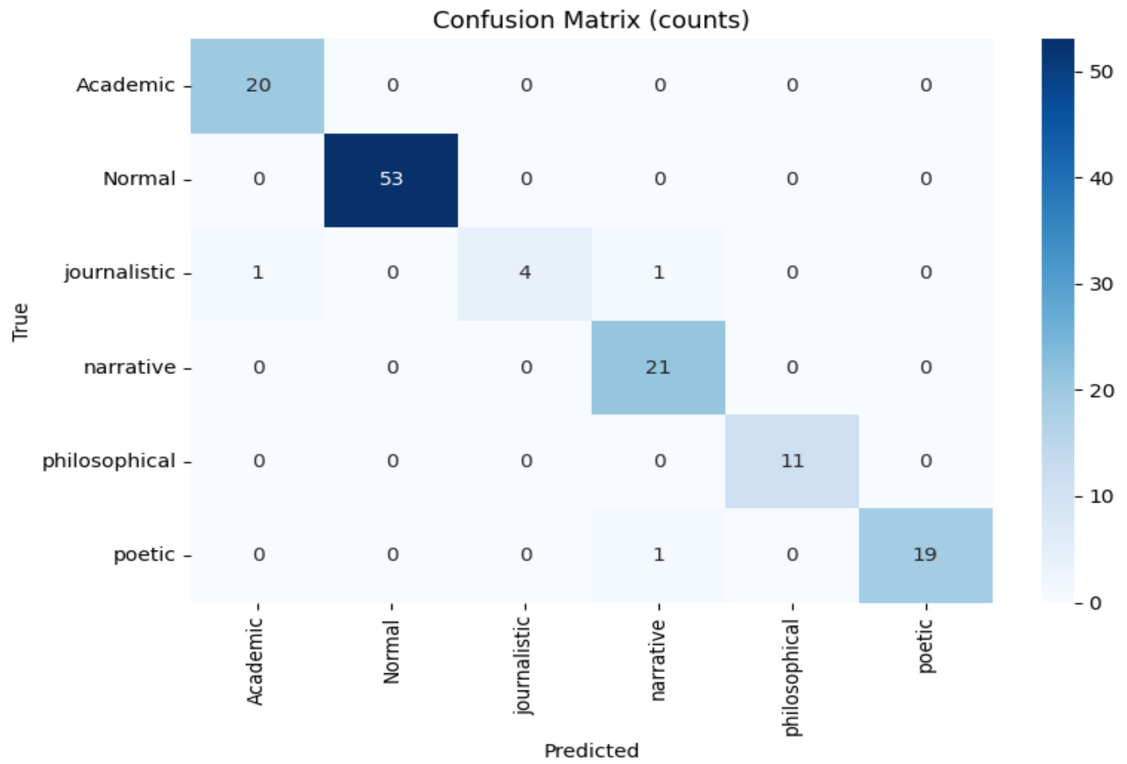
Balanced across genres

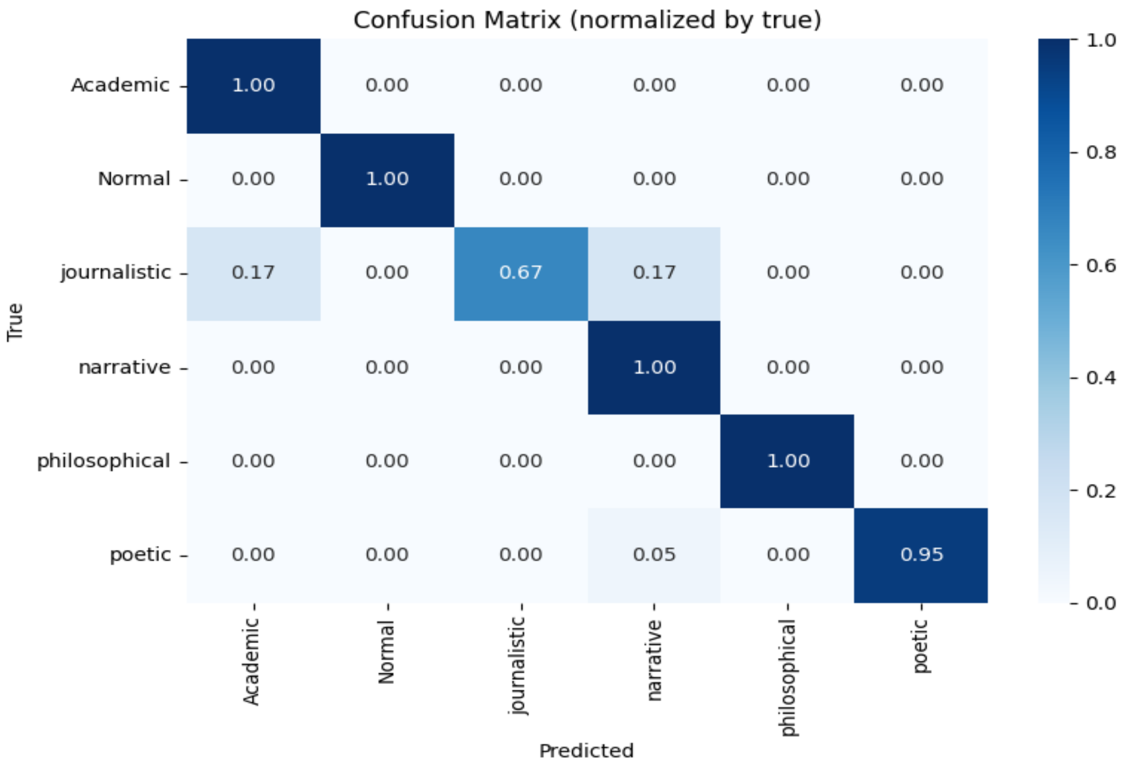
98%

Weighted F1

Overall effectiveness

Key insight: Strong performance across all genres with clear style differentiation. The model successfully distinguishes between writing styles independent of content.





Confusion Matrix Analysis

✅ Strong Diagonal

High correct predictions across all genres demonstrate robust classification.

✅ Clear Separation

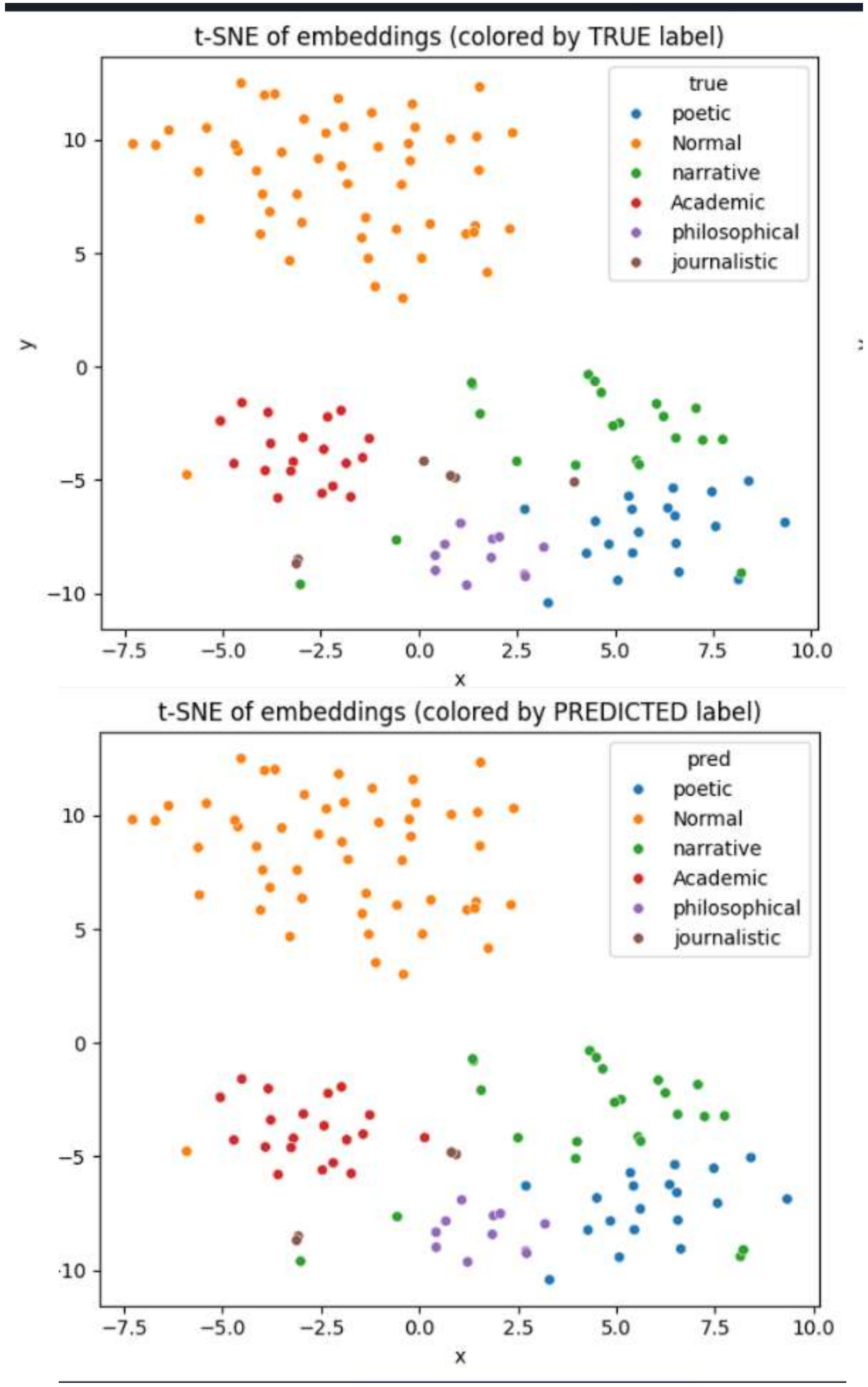
Poetic, Academic, and Philosophical styles show distinct signatures.

⚠️ Expected Overlap

Some confusion between Normal and other genres—Normal is a catch-all category.

⚠️ Minor Confusion

Slight overlap between Narrative and Journalistic reflects natural genre boundaries.



t-SNE Visualisation: Style Clusters

Distinct Clusters

- Poetic: Tight, well-separated cluster
- Academic: Compact, distinct grouping
- Philosophical: Clear separation
- Narrative: Moderate separation

Overlapping Regions

Normal spreads across space, whilst Journalistic mixes with narrative styles.

Model success: Predicted plot mirrors true labels almost perfectly, showing learned style representations.

Example Predictions

Correctly Classified: Poetic

"In the evening, she whispered to the moon about the sea of memories."

Predicted: Poetic (98.5% confidence) ✓

Why? High quote frequency, metaphorical language, figurative expressions.

Most Important Features by Genre

Poetic

High dialogue markers, first-person pronouns, complex readability

Academic

Low dialogue, high passive voice, formal punctuation

Journalistic

Balanced metrics, moderate complexity, factual tone

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Team Contribution

G.Yeshwanth Ram – Contribution in creating dataset and labeled the dataset and implemented the stylometric features. Created the GIT Repo. And presentation.

A.Saketh Krishna – Contribution in creating the Entire pipeline and validation with test dataset. And helped in presentation

B.Adrija – Deployed the model in hugging face with the weights and created the the GUI for website and Helped in pipeline implementation

L.Mohan Royal – Helped in training the model with python and he implemented preprocessing. And helped in presentation.

V.Dheeraj Sai Sudheesh – Helped in testing the model and deployment of the final copy .Helped in presentation.

Key Takeaways

Hybrid Approach Works

Combining stylometric features with semantic embeddings achieves robust style classification independent of content.

Clear Style Signatures

Each genre exhibits distinct linguistic patterns that the model successfully learns and differentiates.

Practical Applications

This research enables authorship attribution, educational feedback, literary analysis, and intelligent content recommendation systems.